Growth and decay of currents in circuits with resistance, capacitance and inductance (Helmholtz's equations).

12.1 GROWTH OF CURRENT IN A CIRCUIT CONTAINING A RESISTANCE AND INDUCTANCE

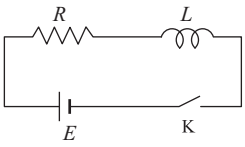


FIG. 12.1

Consider a circuit having an inductance L and a resistance R connected in series to a cell of steady emf E (Fig. 12.1). When the key K is pressed, there is a gradual growth of current in the circuit from zero to maximum value I_0. Let I be the instantaneous current at any instant.

Then, the induced back emf $\varepsilon = -L\dfrac{dI}{dt}$

$$\therefore \quad E = RI + L\frac{dI}{dt} \qquad \ldots(1)$$

When the current reaches the maximum value I_0, the back emf,

$$L\frac{dI}{dt} = 0$$

Hence $$E = RI_0 \qquad \ldots(2)$$

Substituting this value for E in Eq. (1),

$$RI_0 = RI + L\frac{dI}{dt} \text{ or } R(I_0 - I) = L\frac{dI}{dt}$$

or $$\frac{dI}{I_0 - I} = \frac{R}{L}dt$$

Integrating, $$-\log (I_0 - I) = \frac{R}{L}t + C \qquad \ldots(3)$$

where C is the constant of integration.

When $t = 0$, $I = 0$, $\quad \therefore -\log_e I_0 = C$

Substituting this value of C in Eq. (3),

$$-\log (I_0 - I) = \frac{R}{L}t - \log_e I_0 \quad \text{or} \quad \log_e (I_0 - I) - \log_e I_0 = -\frac{R}{L}t$$

$$\log_e \frac{(I_0 - I)}{I_0} = -\frac{R}{L}t$$

$$\frac{I_0 - I}{I_0} = e^{-(R/L)t} \text{ or } 1 - \frac{I}{I_0} = e^{-(R/L)t}$$

$$\therefore \quad I = I_0 (1 - e^{-(R/L)t}) \qquad \ldots(4)$$

Eq. (4) gives the value of the instantaneous current in the LR circuit. The quantity (L/R) is called the *time constant* of the circuit.

$$\text{If } \frac{L}{R} = t,\ I = I_0(1 - e^{-1}) = I_0\left(1 - \frac{1}{e}\right) = 0.632 I_0$$

Thus, *the time constant L/R of a L-R circuit is the time taken by the current to grow from zero to 0.632 times the steady maximum value of current in the circuit.*

Similarly, when $t = 2L/R$, $3L/R$ …, the value of current will be 0.8647, 0.9502, … of the final maximum current.

When $t = 0$, $I = 0$ and when $t \to \infty$, $I = I_0$.

Greater the value of L/R, longer is the time taken by the current I to reach its maximum value (Fig. 12.2).

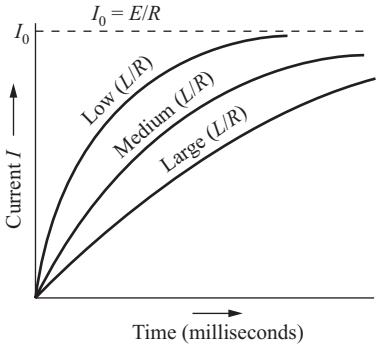


FIG. 12.2

12.2 DECAY OF CURRENT IN A CIRCUIT CONTAINING L AND R

When the circuit is broken, an induced emf, equal to $-L\dfrac{dI}{dt}$ is again produced in the inductance L and it slows down the rate of decay of the current. The current in the circuit decays from maximum value I_0 to zero. During the decay, let I be the current at time t. In this case $E = 0$. The emf equation for the decay of current is

$$0 = RI + L\frac{dI}{dt} \qquad \ldots(1)$$

$$\therefore \quad \frac{dI}{I} = -\frac{R}{L}dt$$

Integrating, $\log_e I = -\dfrac{R}{L}t + C$ where C is a constant

When $t = 0$, $I = I_0$ $\quad \therefore \quad \log_e I_0 = C$

$$\log_e I = -\frac{R}{L}t + \log_e I_0, \quad \text{or} \quad \log_e \frac{I}{I_0} = -\frac{R}{L}t$$

$$\frac{I}{I_0} = e^{-(R/L)t}$$

$$\therefore \quad I = I_0 e^{-(R/L)t} \qquad \ldots(2)$$

Eq. (2) represents the current at any instant t during decay. A graph between current and time is shown in Fig. 12.3.

When $t = L/R$, $I = I_0 e^{-1} = \dfrac{1}{e} I_0 = 0.365\, I_0$

$t = 2L/R$, $I = I_0 e^{-2} = 0.1035\, I_0$

$t = 3L/R$, $I = I_0 e^{-3} = 0.05\, I_0$

Therefore, *the time constant L/R of a R–L circuit may also be defined as the time in which the current in the circuit falls to 1/e of its maximum value* when external source of e.m.f. is removed.

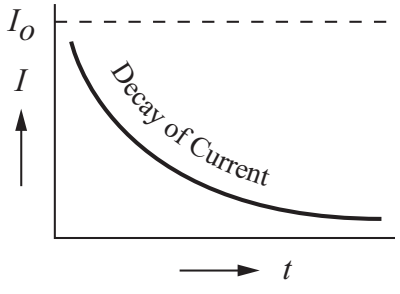


FIG. 12.3

(*i*) When $t = 0.05$ *sec*., $I = 2e^{-10 \times 0.05} = 1.213\,A$

(*ii*) When $t = 0.2$ *sec*., $I = 2e^{-10 \times 0.2} = 0.271\,A$

12.3 GROWTH AND DECAY OF CHARGE IN A CIRCUIT CONTAINING RESISTANCE AND CAPACITOR (CHARGING AND DISCHARGING OF A CAPACITOR THROUGH A RESISTOR)

(*a*) Growth of Charge. A capacitor C and a resistance R are connected to a cell of emf E through a Morse key K (Fig. 12.6). When the key is pressed, a momentary current I flows through R. At any instant t, let Q be the charge on the capacitor of capacitance C.

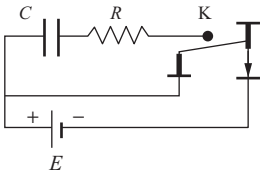


FIG. 12.6

P.D. across capacitor = Q/C

P.D. across resistor = RI

The emf equation of the circuit is

$$E = (Q/C) + RI \qquad \ldots(1)$$

$$E = (Q/C) + R\,(dQ/dt) \qquad \because \quad I = dQ/dt$$

The capacitor continues getting charged till it attains the maximum charge Q_0. At that instant $I = dQ/dt = 0$. The P.D. across the capacitor is $E = Q_0/C$.

i.e., when, $$Q = Q_0, \frac{dQ}{dt} = 0 \text{ and } E = \frac{Q_0}{C}$$

$\therefore$ $$\frac{Q_0}{C} = \frac{Q}{C} + R\frac{dQ}{dt}$$

$$(Q_0 - Q) = CR\frac{dQ}{dt}$$

$$\left(\frac{dQ}{Q_0 - Q}\right) = \frac{dt}{CR} \qquad \ldots(2)$$

Integrating, $-\log_e (Q_0 - Q) = \frac{t}{CR} + K$

where K is a constant.

When $t = 0, Q = 0 \quad \therefore \quad -\log_e Q_0 = K$

$$-\log_e (Q_0 - Q) = \frac{t}{CR} - \log_e Q_0$$

$$\log_e (Q_0 - Q) = -\frac{t}{CR} + \log_e Q_0$$

$$\log_e (Q_0 - Q) - \log_e Q_0 = -\frac{t}{CR}$$

$$\log_e \left(\frac{Q_0 - Q}{Q_0}\right) = -\frac{t}{CR}$$

$$\frac{Q_0 - Q}{Q_0} = e^{-\frac{t}{CR}} \text{ or } 1 - \frac{Q}{Q_0} = e^{-\frac{t}{CR}}$$

$$\therefore \quad Q = Q_0\left(1 - e^{-\frac{t}{CR}}\right) \qquad ...(3)$$

The term CR is called time constant of the circuit.

At the end of time $t = CR, Q = Q_0\,(1 - e^{-1}) = 0.632\,Q_0$

Thus, *the time constant may be defined as the time taken by the capacitor to get charged to 0.632 times its maximum value.*

The growth of charge is shown in Fig. 12.7.

The rate of growth of charge is

$$\frac{dQ}{dt} = \frac{Q_0}{CR}\,e^{-\frac{t}{CR}} = \frac{I}{CR}\,(Q_0 - Q)$$

Thus it is seen that smaller the product CR, the more rapidly does the charge grow on the capacitor.

The rate of growth of the charge is rapid in the beginning and it becomes less and less as the charge approaches nearer and nearer the steady value.

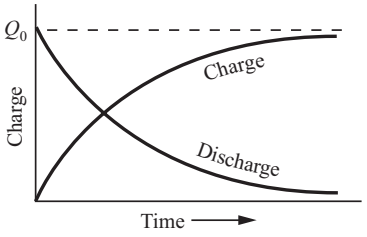


FIG. 12.7

(*b*) Decay of charge (Discharging of a Capacitor through Resistance)

Let the capacitor having charge Q_0 be now discharged by releasing the Morse key K (Fig. 12.6). The charge flows out of the capacitor and this constitutes a current. In this case $E = 0$.

$$R\,\frac{dQ}{dt} + \frac{Q}{C} = 0 \qquad \ldots(1)$$

or $$\frac{dQ}{Q} = -\frac{1}{CR}\,dt$$

Integrating, $\log_e Q = -\dfrac{t}{CR} + K,$ where K is a constant

When $t = 0, Q = Q_0 \therefore \log_e Q_0 = K$

$$\log_e Q = -\frac{t}{CR} + \log_e Q_0$$

or $$\log_e \frac{Q}{Q_0} = -\frac{t}{CR} \quad \text{or} \quad \frac{Q}{Q_0} = e^{-t/CR}$$

$$\therefore \quad Q = Q_0 e^{-t/CR} \qquad \ldots(2)$$

This shows that the charge in the capacitor decays exponentially and becomes zero after infinite interval of time (Fig. 12.7).

The rate of discharge is

$$I = \frac{dQ}{dt} = -\frac{Q_0}{CR}\,e^{-t/CR} = -\frac{Q}{CR} \qquad \ldots(3)$$

Thus, smaller the time-constant CR, the quicker is the discharge of the capacitor.

In Eq. (2), if we put $t = CR$, then $Q = Q_0 e^{-1} = 0.368\,Q_0$.

Hence *time constant may also be defined as the time taken by the current to fall from maximum to 0.368 of its maximum value.*

Example 1. *A capacitor is charged by DC supply through a resistance of 2 megaohms. If it takes 0.5 seconds for the charge to reach three quarters of its final value, what is the capacitance of the capacitor?*

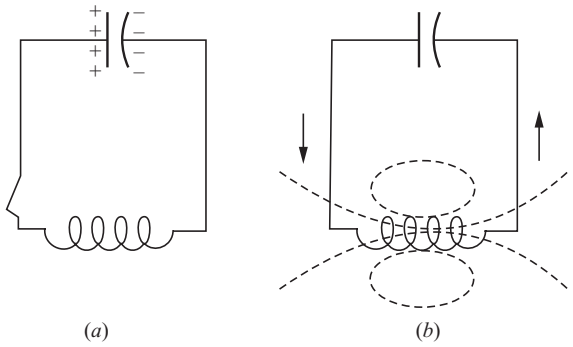


FIG. 12.12

- When the resistance of the circuit is low, a large number of oscillations take place per second. The changes in electric and magnetic fields give rise to electromagnetic radiations, which are propagated through space with the velocity of light.
  The frequency of these oscillations is given by

$$v = \frac{1}{2\pi\sqrt{LC}}.$$

- These electromagnetic oscillations form the basis of wireless telegraphy. With the help of codes, the messages can be transmitted from one place to another.

12.6 GROWTH OF CHARGE IN A CIRCUIT WITH INDUCTANCE, CAPACITANCE AND RESISTANCE

Consider a circuit containing an inductance L, capacitance C and resistance R joined in series to a cell of emf E (Fig. 12.13). When the key K is pressed, the capacitor is charged. Let Q be the charge on the capacitor and I the current in the circuit at an instant t during charging. Then, the p.d. across the capacitor is Q/C and the self induced emf in the inductance coil is $L\,(dI/dt)$, both being opposite to the direction of E. The p.d. across the resistance R is RI.

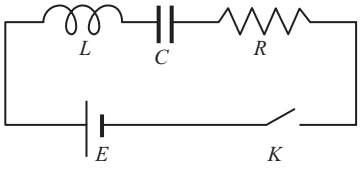


FIG. 12.13

The equation of emf's is

$$L\frac{dI}{dt} + RI + \frac{Q}{C} = E \qquad \ldots(1)$$

But $$I = \frac{dQ}{dt} \text{ and } \frac{dI}{dt} = \frac{d^2Q}{dt^2}$$

$\therefore$ $$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = E$$

or $$\frac{d^2Q}{dt^2} + \frac{R}{L}\frac{dQ}{dt} + \frac{Q - CE}{LC} = 0$$

Putting $\frac{R}{L} = 2b$ and $\frac{1}{LC} = k^2$, we have

$$\frac{d^2Q}{dt^2} + 2b\frac{dQ}{dt} + k^2\,(Q - CE) = 0 \qquad \ldots(2)$$

Let $x = Q - CE$. Then $\frac{dx}{dt} = \frac{dQ}{dt}$ and $\frac{d^2x}{dt^2} = \frac{d^2Q}{dt^2}$

Eq. (2) becomes, $\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + k^2x = 0$ …(3)

Hence the most general solution of Eq. (3) is

$$x = Ae^{[-b+\sqrt{(b^2-k^2)}]t} + Be^{[-b-\sqrt{(b^2-k^2)}]t}$$

Now, $CE = Q_0$= final steady charge on the capacitor.

$\therefore$ $x = Q - CE = Q - Q_0$

Hence $Q - Q_0 = Ae^{[-b+\sqrt{(b^2-k^2)}]t} + Be^{[-b-\sqrt{(b^2-k^2)}]t}$

or $Q = Q_0 + Ae^{[-b+\sqrt{(b^2-k^2)}]t} + Be^{[-b-\sqrt{(b^2-k^2)}]t}$ …(4)

Using initial conditions:

at $t = 0, Q = 0$

$\therefore$ $0 = Q_0 + (A + B)$ or $A + B = -Q_0$ …(5)

$$\frac{dQ}{dt} = A(-b+\sqrt{(b^2-k^2)})\,e^{[-b+\sqrt{(b^2-k^2)}]t} + B[-b-\sqrt{(b^2-k^2)}]\,e^{[-b-\sqrt{(b^2-k^2)}]t}$$

At $t = 0, \frac{dQ}{dt} = 0$

$$0 = A[-b+\sqrt{(b^2-k^2)}] + B[-b-\sqrt{(b^2-k^2)}]$$

$$\sqrt{(b^2-k^2)}\,[A-B] = b(A+B) = -bQ_0$$

or $A - B = -\frac{Q_0 b}{\sqrt{(b^2-k^2)}}$ …(6)

Solving Eqs. (5) and (6),

$$A = -\frac{1}{2}Q_0\left(1+\frac{b}{\sqrt{(b^2-k^2)}}\right) \quad \text{…(7)}$$

$$B = -\frac{1}{2}Q_0\left(1-\frac{b}{\sqrt{(b^2-k^2)}}\right) \quad \text{…(8)}$$

Substituting the values of A and B in Eq. (4), we have

$$Q = Q_0 - \frac{1}{2}Q_0 e^{-bt}\left[\left(1+\frac{b}{\sqrt{(b^2-k^2)}}\right)e^{\sqrt{(b^2-k^2)}\,t} + \left(1-\frac{b}{\sqrt{(b^2-k^2)}}\right)e^{-\sqrt{(b^2-k^2)}\,t}\right] \quad \text{…(9)}$$

Case I. If $b^2 > k^2$, $\sqrt{(b^2-k^2)}$ is real. The charge on the capacitor grows exponentially with time and attains the maximum value Q_0 asymptotically, (curve 1 of Fig. 12.14). The charge is known as *over damped or dead beat*.

Case II. If $b^2 = k^2$, the charge rises to the maximum value Q_0 in a short time (curve 2 of Fig. 12.14). Such a charge is called *critically damped*.

Case III. If $b^2 < k^2$, $\sqrt{(b^2-k^2)}$ is imaginary.

Let $\sqrt{(b^2-k^2)} = i\omega$ where $i = \sqrt{-1}$ and $\omega = \sqrt{k^2-b^2}$